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5.1 Accidents, Their Causes, Types, Results and Control :

5.1.1 Definition of accident :

"Without paying attention towards what will be its results, the acts of variable working gets implemented unknowingly, suddenly and hastily leading to the injuries to workers and damage to properties is known as an accident."

The unimagined calamity arriving and causing unexpected losses to the machines, tools, workers and other equipments is called an accident. The accident is an unfortunate happening and causes the damages to the industry.

The unintentional acts and anything happens without deliberately in the plant during working which is causing the injuries are known as an accident. The injuries may be temporary or permanent in nature but they leads the condition of workers unsuited for working.

5.1.2 Causes of an accident:

5.1.2.1 Supervisory programme of safety :

1. Insufficient implementation and passing of safety suggestions & instructions.
2. Proper rules for safety are not imposed.
3. The planning of safety is not done as a part of work.
4. The workers support is not obtained to bring safety.

5. Safety appliances are not provided.
6. Improvement is not done in hazardous conditions & atmosphere.

5.1.2.2 Mental conditions of workers :

1. Insufficient awareness towards safety.
2. Lack of co-operation
3. Improper behaviour
4. Inattentiveness or distracted
5. Physically unfit for the work
6. Poor or incapable to control his feelings
7. Mental weakness
8. Emotional or sentimental by nature.

5.1.2.3 Physical conditions of worker :

1. Too much tired, fatigued or exhausted.
2. Deaf or hard of hearing.
3. Physically improper or unworthy for the work.
4. Poor or weak hearted
5. Poor-eyesight
6. Physically handicapped.

5.1.2.4 Unsafe acts :

1. No guards or broken or damaged guards of machines.
2. Working on machine without using safety appliances.
3. Carelessness in maintaining required work condition.
4. Using the machine in overload conditions.
5. To arrange the materials & parts in hazardous manner
6. Insufficient maintenance of workplace.

5.1.2.5 Unsafe conditions :

1. Ineffectiveness of the safety appliances
2. Inavailability of safety appliances
3. Poor maintenance
4. Faulty conditions of machines & tools
5. Improper clothing used by worker as dress, clothes or garments.
6. Insufficient lighting
7. Poor ventilation.
8. Noisy atmosphere.

5.1.3 Types of accidents :

The accidents are classified according to the seriousness of the injuries and the incapability caused in physical condition of the worker to carry out the work. The accident

causing death or not. Also the amount of damages and losses caused to the industry on basis for classifying the accidents. The accidents are classified in four types as follows:

1. Fatal accident
2. Serious accident
3. Minor accident
4. No injury accident

5.1.3.1 Fatal accidents :

The small or big accident in which the loss of life has been occurred then the accident is known as "Fatal accident". The family of person expired in an accident is placed in difficult conditions. The claims for compensation are arises against the industry. By paying the compensation amount the damage is occurred to the economical condition of the industry. Over and above industry is losing an experienced worker. To fill up the place of deceased person industry has to appoint a new man and to make expenses towards his training.

The peoples linked with feeling is the definition of the organization known as a family. The whole family of the expired worker, therefore, suffers from pain, anguish and affliction. Therefore accident can be said as a social calamity. Hence the fatal accident causes the social and economical loss and injuries to the human society.

5.1.3.2 Serious accident :

The accident which causes serious type of physical shortcomings and huge loss to the industrial property is known as serious accident. After taking costlier treatment for a long period of time, the injury caused by this accident left one or other type of physical disability permanently, which reduces the personality of the worker. The great loss created to the machines & equipments due to fire or explosion is also called a serious accident. When greater loss takes place to the industrial building than also the accident is classified as serious accident. As a result of serious accident expenses are to be incurred for medical treatment & compensation to worker as well as repairing cost of building, machines & equipments will be also high. Loss of deficiency of skilled worker experienced by the industry and the time required to fill up that loss reduces the production quality resulting in the increased wastages is creating the further loss to the industry.

5.1.3.3 Minor accident :

The accident which causes minor injuries to the workers and minor damages to the industrial properties like machines, equipments, tools and building is known as minor accident. As a result of this the accident the expenses required to be incurred for medical treatments, payment of compensations, repairing of machines and building are less. In such accidents minor injuries takes place due to falling, slipping and striking against. To cure such minor injuries few hours or days are needed. After minor treatment the worker returns to work. When the parts breakages and explosion causes minor damages to the machines and

building then that incidents are called minor accident. Here in this case with little expenses the machine or building can be repaired and the work of production can be restarted speedily.

5.1.3.4 No injury accident :

The occurrence of the serious or minor accidents which does not result into the injuries than that accident is called no-injury accident. To decide this type of accident more or less damages and loss occurred is not taken into considerations. The attention is not given towards the incurred loss but the type is only decided on the basic of injuries. Therefore it can be said that when there is no injuries of any kind to anybody that accident occurred is called no-injury accident.

5.1.4 Results of an accident :

Accidents are causing mental, physical, economical and social effects. Due to the occurrence of accidents, the plant workers are experiences terror and fear, laxity and slackness, plus fury & excitement. Then they always work under the fear and terror in a frightened state of mind. This results into the displeasure & insecurity type of social effect. It causes its bad effects by spoiling the familiar atmosphere of victim. The family members of the victim worker under the frightened state of mind may feel sick but if the worker loss his life in accident then the condition of his family becomes worst. If such accidents occurs often and often then it results into the loss of morale of the workers.

The workers injured in accident has to undergo necessary medical treatment and during the period of medical treatment he will not return for the work. Therefore his work has to be assigned to the other worker. The quantity and quality of production in such cases will be reduced, due to that wastage will increase and in proportion to this increased wastage, the cost of production also increases. The economical condition of the industry is affected by the decreased margin of profit. The further reduction in economical condition is caused by the expenses made for treatment, paying of compensation, and cost incurred for repairing the machines, equipments, tools and building etc. Over and above these direct losses, the indirect losses like the wastages of production time of workers and supervisors has to be faced by the industry thus additional financial liabilities are arising. Some of the results of an accident are as follows :

1. Harrassment, tyranny, tedium and boredom is created due to the accident.
2. The production stops.
3. The quality of products reduces.
4. The wastages of production increases.
5. Minor injuries are taking place.
6. Some injuries may lead to the permanent disability.
7. Loss of life may occur due to an accident.
8. Loss and damages of the industrial property is caused.
9. Expenses for medical treatment has to be incurred.

10. Claim for workers compensation arises.
11. Additional financial liability of paying the compensation has to be faced by industries.
12. Damages are caused to the machines, tools, equipments & buildings.
13. Frighten atmosphere results in an industry due to frequent accidents.
14. The family members of the victim worker also experience the fear which affects their mental condition.
15. The destruction of a family caused by the death of the worker, who is the support of their earnings.
16. Labour turnover increase in unsafe industries.
17. The target of production cannot be obtained.
18. Timely dispatches of production to the customers cannot be maintained.
19. Loss of customers and market.
20. The arrangement for finding new market has to be made.
21. The customers once lost may get habituated of alternative brands, due to that they will not return back even in future.
22. The pressing of workers trade unions increases..
23. Accident creates strikes and lockouts.
24. Management has to think for the additional works related to restoration of safety and solving the labour problems.
25. In the circumstances in which if the morale of management breaks then managing an industry becomes difficult.

5.1.5 Control of an accident :

The factors related to control the accidents are as follows :

1. Improvement in safety supervision.
2. Improvement in mental condition of workers.
3. Improvement in physical condition of workers.
4. Improvement & addition of safety devices.
5. Know-how of workers for safety devices.
6. Celebration of safety week and month.

5.1.5.1 Improvement in safety supervision :

The condition of machines should be properly maintained and then machines are used so that there will not be any injury to workers. Proper synthesis should be obtained between the technical skill and safety. The workers should be trained to work safely. The safety rules should be laid down and implemented strictly. The workers contribution towards safety should be decided. At the time of planning each job, the hazards present it should be thought of as regards to their type as well as the steps to prevent the occurrence of an accident. The

work should be done with awareness towards safety. The rules for safety should be imposed and followed strictly but with the due co-operation of the workers, to make the efforts related to safety very effective.

5.1.5.2 Improvement in mental condition of workers :

It is necessary to give priority and importance to the good mental conditions of the workers for preventing accidents. When the workers are working with their poor mental conditions, then they are found responsible for accident. Fails to control their feelings, inattentiveness, non-cooperative, hasty by nature and those are taking more time to understand the circumstances leading to accident are some of the poor mental conditions of the workers. Control of feeling, attentiveness towards the work, work with co-operation, don't show any hastiness, and know the alterations in machine immediately are some of the important mental conditions for preventing the accidents.

5.1.5.3 Improvement in physical condition of workers :

The workers those are working within their physical capacity and take rest till they experience fatigue are preventing accidents. The workers those are deaf, with poor eye-sight and physically handicapped are casually causing the accidents. By using the aids to correct deafness and eye-sight worker can remain safe against the accidents caused by physical short-comings. The physically handicapped workers if discharges their duties within their physical capacity can save them against accidents. Sufficient nourishment of the worker is most important for their physical health. This becomes possible when the workers are paid with good wages and salaries. The accidents can be prevented by working with physical and mental health.

5.1.5.4 Improvement and addition of safety devices :

The safety devices such as welding glass, goggles for gas welding and grinding should be in their good conditions. The safety devices must be kept ready in sufficient numbers. The machinery guards should be strong and well maintained. Readiness of fire fighting equipments should be maintained and sufficient number of such equipments are to be kept ready at accessible locations. The insulated gloves, tools, rubber mat, insulated stands are to be provided in sufficient numbers. Such devices must be used at time of working is very very important.

5.1.5.5 Knowhow of workers for safety devices :

The different safety devices can be used properly if the workers are trained to develop their knowhow for the type, construction and proper use of the safety devices. Workers' knowhow can only prevent accidents using safety devices. The workers trained for preventing accidents and fire are knowing the ways of preventing the accidents, therefore, they will work cautiously, vigilantly and with greater awareness and alertness. Hence they are definitely reduces the accidents by their safe way of working. Therefore sufficient training for safety devices, their use, methods of fighting fire are to be given for the prevention of accidents.

5.1.5.6 Celebration of safety week and month :

This type of celebration are the experiments which has proved successful to increase the awareness of accidents & safety. The workers participated in such functions are found improving their behaviour and working with more enthusiasm. They starts realizing their responsibilities. Their feelings can be awakened to work with vigilant and conscious mind. The work practices adopted in their improved state of vigilance can definitely prevent the accident and brings safety in the plant and industries.

5.2 Mechanical and Electrical Hazards :

5.2.1 Mechanical hazards :

Accidents in machines are generally occurs due to the gearing, belts, pulleys and shaftings. The injury can occur even during the material transportation and using hand tools. By striking against a machine, entrapping cloths, hands or legs in machine drives, and pieces of materials and chips can also cause the injuries. The mechanical hazards are divided into three main types keeping in mind the above stated circumstances as follows :

1. Material handing hazards
2. Hand tool hazards
3. Machine hazards

5.2.1.1 Material handling hazards :

The reasons for the development of material handling hazards are as follows :

1. Improper stacking of the materials.
2. Carelessness in loading & unloading of materials.
3. Falling of the products while placing in rack, platform and bin.
4. Unsafe condition of crane and slings.
5. Use of material handling equipments having unsafe design.
6. Use of material handling equipments under overload condition.

5.2.1.2 Hand tool hazards :

The reasons for accident occurring due to hand tool hazards are as follows :

1. Poor condition and insufficient maintenance.
2. Insufficient care in using the hand tools.
3. Improper work of maintenance & inspection for hand tools.
4. Difficulties in gripping the hand tools.
5. Mismanaged or disordered working with hand tools.

5.2.1.3 Machine hazards :

1. Improper machine guards
2. Poor condition of the guards
3. Insufficient guards
4. Wrong or faulty location of machine controls

5. Poor or deteriorated machine condition.
6. Working surfaces of the machine are not protected.
7. Careless maintenance of machine.

5.2.2 Electrical Hazards :

Wireman and electricians are climbing on the pole and work with the current carrying conductors carrying high voltages. Their work involves the cutting and joining of the conductors after removing the insulations. Therefore it is necessary for them to work using the insulated tools. The electrical hazards are as follows :

1. Defective construction
2. Improper use
3. Insufficient protective devices
4. Carelessness in line working
5. Portable electrical tools.

5.2.2.1 Faulty construction :

When there is insufficient clearance between two conductors and earthing than faulty construction exists. Insufficient guards, insufficient devices to maintain & supply the electricity & improper earthing of electrical poles leads to the hazards due to faulty construction.

5.2.2.2 Improper use :

If the rules for using the electrical equipments and instructions related to it are not followed then this type of hazards arises. The particular guidelines and instructions are given for using the electrical equipments but at the time of working due care is not exercised to follow the laid down instructions and hence this type of hazards have to be faced.

5.2.2.3 Insufficient safety equipments :

Rubber gloves, insulated pliers & other tools, safety belt, insulated stand and shoes are not kept in stock in sufficient numbers. If each safety device is available in stock but not utilized then this type of hazard arises. To remain safe against the electrical hazards like touching the live conductors, short circuiting, accidental grounding, breakage of overload connections etc. it is necessary to have safety devices & tools in sufficient number and in good conditions as well as utilize them while working is necessary. Therefore this working need should be satisfied to prevent the accidents due to electrical hazards.

5.2.2.4 Carelessness in line work :

Before starting the work if sufficient line clearance is not kept exercising due care, then it can lead to the accidents due to electrical hazards, such line work is always unsafe.

5.2.2.5 Portable electrical tool :

For the fabrication of steel structures, boiler and tanks as well as in body fabrication work of an automobile the extensive use of portable hand drilling machines is very common.

When working with such machine, the person will be in contact of ground. If such hand-tool body or frame is not ground then accident occurs due to electric shock. Hence by using the three wire code arrangements must be made to provide safety while making the electrical connections to the electrical hand tools. If the electrical tool is connected to the ground with a clamp then it can provide more safety.

5.2.3 Reasons for electrical shock :

1. To hold or catch the phase wire by mistake.
2. Earthing is not done or wrong earthing.
3. Material of wire is of poor & inferior quality.
4. The ampere-capacity of the wire is less.
5. Wiring maintenance is not done.
6. Due to the worn out insulation conductors are in its bare condition.
7. Electrical work carried out by unauthorised persons.
8. Safety switches for preventing the accident due to short circuiting is not used.

5.2.4 Remedies for preventing mechanical hazards :

1. Material handling should be done properly.
2. Provide necessary support to the materials.
3. Keep equipments in good condition.
4. Protect the hands by using hand gloves & legs by shoes.
5. Maintain the flooring in good & cleaned condition.
6. Arrange the material properly.
7. Use always right types of hand tools.
8. Use hand tools of good condition.
9. Use the hand tools within its capacity.
10. Maintain the hand tools with sufficient care.
11. Acquire sufficient grip while using the hand tools.
12. Provide particular type of guard providing sufficient guarding for pulley, belt, shaft, coupling and gears.
13. Use the guards for starting and stopping levels.
14. Keep the prime move safe by providing guards to it.
15. Decide the chip removal to avoid injuries during machining.
16. Handle the material manually within the limit of 50 kg.
17. Do not stand below the weight lifted by crane or hoist.
18. Every year emphasis complete check up material handling equipments.
19. Check the wire ropes in every ten days, which are used for lifting the load.
20. Always write load capacity on material handling equipment.

2.5 Remedies for preventing electrical hazards :

1. Do not guess for the circuit whether it is continuing or closed but believe that it is supplying the power.
2. Check the circuit using proper tools & equipments.
3. Make sure that the current is not flowing through the wire before touching it.
4. Use rubber gloves, insulated plier, screw driver and fuse tongs.
5. After moving sufficient away from the line start the supply.
6. Implement the instructions of danger and provide fencing around the dangerous location.
7. Make temporary connections even safe.
8. Use of electrical equipments should be done only by the authorised person.
9. Check the conductors and insulation for safety.
10. Do not use fuse with wire looping.
11. Do not carry out work on live lines only when you are alone.
12. Do not check the power using a lamp but test the voltage of the line.
13. Reduce the proportionate use of aluminium conductors.
14. Fix the line wires at sufficient height.
15. To avoid the gathering of two conductors, keep sufficient clearance.
16. Start repairing after disconnecting the supply and keep it disconnected till the work completes.
17. Check for electrical safety & thereafter supply the power.
18. Use the electrical hand tools within their capacities.
19. Use the equipments on a work for which they are meant.
20. Use the hand tools after giving them earth connection.
21. Use rubber insulated gloves while working with electrical hand tools.

5.3 Describe salient points of Factories Act 1948 for health & safety wash room :**• HEALTH :**

Cleanliness : (1) Every factory shall be kept clean and free from effluvia arising from any drain, privy or other nuisance, and in particular :

- (a) Accumulation of dirt and refuse shall be removed daily by sweeping or by any other effective method from the floors and benches of work-rooms and from staircases and passages, and disposed of in a suitable manner;
- (b) The floor of every workroom shall be cleaned at least once in every week by washing, using disinfectant, where necessary, or by some other effective method;
- (c) Where a floor is liable to become wet in the course of any manufacturing process to such extent as is capable of being drained, effective means of drainage shall be provided and maintained;

- (d) All inside walls and partitions, all ceilings or tops of rooms and all walls, sides and tops of passages and staircases shall -
- (i) where they are 1* (painted otherwise than with washable water-paint) or varnished, be repainted or revarnished at least once in every period of five years;
 - 2* [(ia) where they are painted with washable water paint, be repainted with at least one coat of such paint at least once in every period of three years and washed at least once in every period of six months;]
 - (ii) where they are painted or varnished or where they have smooth impervious surfaces, be cleaned at least once in every period of fourteen months by any method as may be prescribed;
 - (iii) In any other case, be kept whitewashed or colourwashed, and the whitewashing or colourwashing shall be carried out at least once in every period of fourteen months;
 - 2* [(dd) all doors and window frames and other wooden or metallic framework and shutters shall be kept painted or varnished and the painting or varnishing shall be carried out at least once in every period of five years;]

(2) If, in view of the nature of the operations carried on 1* (in a factory or class or description of factories or any part of a factory or class or description of factories), it is not possible for the occupier to comply with all or any of the provisions of subsection (1), the State Government may be order exempt such factory or class or description of factories (or part) from any of the provisions of that sub-section and specify alternative methods of keeping the factory in a clean state.

• **Disposal of wastes and effluents :**

Disposal of wastes and effluents. 3* [(1) Effective arrangements shall be made in every factory for the treatment of wastes and effluents due to the manufacturing process carried on therein, so as to render them innocuous and for their disposal.]

(2) The State Government may make rules prescribing the arrangements to be made under sub-section (1) or requiring that the arrangements made in accordance with subsection (1) shall be approved by such authority as may be prescribed.

• **Ventilation and temperature :**

(1) Effective and suitable provision shall be made in every factory for securing and maintaining in every workroom -

- (a) Adequate ventilation by the circulation of fresh air, and

1. Subs. by Act 94 of 1976, s. 8, for the word "painted" (w.e.f. 26-10-1976)
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2. Ins. by s. 8, *ibid.*, (w.e.f. 26-10-1976).
3. Subs. by s. 9, *ibid.*, for sub-section (1) (w.e.f. 26-10-1976).

- (b) Such a temperature as will secure to workers therein reasonable conditions of comfort and prevent injury to health;—
and in particular, —
- (i) walls and roofs shall be of such material and so designed that such temperature shall not be exceeded but kept as low as practicable;
- (ii) where the nature of the work carried on in the factory involves, or is likely to involve, the production of excessively high temperatures, such adequate measures as are practicable shall be taken to protect the workers there from, by separating the process which produces such temperatures from the workroom, by insulating the hot parts or by other effective means.
- (3) If it appears to the Chief Inspector excessively high temperatures in any factory can be reduced by the adoption of suitable measures, he may, without prejudice to the rules made under subsection (2), serve on the occupier, an order in writing specifying the measures which, in his opinion, should be adopted, and requiring them to be carried out before a specified date.]

Dust and fume :

- (1) In every factory in which, by reason of the manufacturing process carried on, there is given off any dust or fume or other impurity of such a nature and to such an extent as is likely to be injurious or offensive to the workers employed therein, or any dust in substantial quantities, effective measures shall be taken to prevent its inhalation and accumulation in any workroom, and if any exhaust appliance is necessary for this purpose, it shall be applied as near as possible to the point of origin of the dust, fume or other impurity, and such point shall be enclosed so far as possible.
- (2) In any factory no stationary internal combustion engine shall be operated unless the exhaust is conducted into the open air, and no other internal combustion engine shall be operated in any room unless effective measures have been taken to prevent such accumulation of fumes therefrom as are likely to be injurious to workers employed in the room.

Artificial humidification :

- (1) In respect of all factories in which the humidity of the air is artificially increased, the State Government may make rules,
- Prescribing standards of humidification;
 - Regulating the methods used for artificially increasing the humidity of the air;
 - Directing prescribe tests for determining the humidity of the air to be correctly carried out and recorded,
 - Prescribing methods to be adopted for securing adequate ventilation and cooling of the air in the workrooms.
- Subs. by Act 20 of 1987, s. 6, (w.e.f. 1-12-1987).

- (2) In any factory in which the humidity of the air is artificially increased, the water used for the purpose shall be taken from a public supply, or other source of drinking water, or shall be effectively purified before it is so used.
 - (3) If it appears to an Inspector that the water used in a factory for increasing humidity is required to be effectively purified under subsection (2) is not effectively purified, he may serve on the manager of the factory an order in writing, specifying the measures which in his opinion should be adopted, and requiring them to be carried out before specified date.
- **Overcrowding :**
 - (1) No room in any factory shall be overcrowded to an extent injurious to the health of the workers employed therein.
 - (2) Without prejudice to the generality of sub-section (1), there shall be in every workroom of factory in existence on the date of the commencement of this Act at least 1* (9.9 cubic meters) and of a factory built after the commencement of this Act at least 1* (14.2 cubic meters) of space for every worker employed therein, and for the purpose of this sub-section no account shall be taken of any space which is more than 1* (4.2 meters) above the level of the floor of the room.
 - (3) If the Chief Inspector by order in writing so requires, there shall be posted in each workroom of a factory a notice specifying the maximum number of workers who may, in compliance with the provisions of this section, be employed in the room.
 - (4) The Chief Inspector may by order in writing exempt, subject to such conditions, if any, as he may think fit to impose, any workroom from the provisions of this section, if he is satisfied that compliance therewith in respect of the rooms is unnecessary in the interest of the health of the workers employed therein.
 - **Lighting :**
 - (1) In every part of a factory where workers are working or passing there shall be provided and maintained sufficient and suitable lighting, natural or artificial, or both.
 - (2) In every factory all glazed windows and skylights used for the lighting of the workrooms shall be kept clean on both the inner and outer surface and, so far as compliance with the provisions of any rules made under sub-section (3) of section 13 will allow, free from obstruction.
 - (3) In every factory effective provision shall, so far as is practicable, be made for the prevention of –
 - (a) glare, either directly from a source of light or by reflection from a smooth or polished surface;

1. Subs. by Act 20 of 1987, s. 7 (w.e.f. 1-12-1987).

- (b) the formation of shadows to such an extent as to cause eye-strain or the risk of accident to any worker.
- (4) The State Government may prescribe standards of sufficient and suitable lighting for factories or for any class of description of factories or for any manufacturing process.

Drinking water :

- (1) In every factory effective arrangements shall be made to provide and maintain at suitable points conveniently situated for all workers employed there in a sufficient supply of wholesome drinking water.
- (2) All such points shall be legibly marked "drinking water" in a language understood by majority of the workers employed in the factory, and no such point shall be situated within 1* (six meters of any washing place, urinal, latrine, spittoons, open drain carrying sullage or effluent or any other source of contamination) unless a shorter distance is approved in writing by the Chief Inspector.
- (3) In every factory wherein more than two hundred and fifty workers are ordinarily employed, provision shall be made for cooling drinking water during hot weather by effective means and for distribution thereof.
- (4) In respect of all factories or any class or description of factories the State Government may make rules for securing compliance with the provisions of sub-sections (1), (2) and (3) and for the examination by prescribed authorities of the supply and distribution of drinking water in factories.

Latrines and urinals :

- (1) In every factory
- (a) Sufficient latrine and urinal accommodation of prescribed types shall be provided conveniently situated and accessible to workers at all times while they at the factory;
- (b) Separate enclosed accommodation shall be provided for male and female workers;
- (c) Such accommodation shall be adequately lighted and ventilated, and no latrine or urinal shall, unless specially exempted in writing by the Chief Inspector, communicate with any workroom except through an intervening open space or ventilated passage;
- (d) All such accommodation shall be maintained in a clean and sanitary condition at all times;
- (e) Sweepers shall be employed whose primary duty it would be to keep clean latrines, urinals and washing places.

1. Subs. by Act 20 of 1987, s. 7 (w.e.f. 1-12-1987).
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2. Subs. by s. 9 *ibid.* (w.e.f. 1-12-1987).

- (2) In every factory wherein more than two hundred and fifty workers are ordinarily employed –
- All latrine and urinal accommodation shall be of prescribe sanitary types;
 - The floors and internal walls, up to a height of 2* (ninety centimetres) of the latrines and urinals and the sanitary blocks shall be laid in glazed tiles or otherwise finished to provide a smooth polished impervious surface;
- (3) The State Government may prescribe the number of latrines and urinals to be provided in any factory in proportion to the numbers of male and female workers ordinarily employed therein, and provide for such further matters in respect of sanitation in factories, including the obligation of workers in this regard, as it considers necessary in the interest of the health of the workers employed therein.

• **Spittoons :**

- In every factory there shall be provided a sufficient number of spittoons in convenient places and they shall be maintained in a clean and hygienic condition.
- The State Government may make rules prescribing the type and the number of spittoons to be provided and their location in any factory and provide for such further matters relating to their maintenance in a clean and hygienic condition.
- No person shall spit within the premises of a factory except in the spittoon provided for the purpose and a notice containing this provision and the penalty for its violation shall be prominently displayed at suitable places in the premises.
- Whoever spits in contravention of sub-section (3) shall be punishable with fine not exceeding five rupees.

SAFETY :

Fencing of machinery :

- In every factory the following, namely,
 - Every moving part of a prime mover and every flywheel connected to a prime mover whether the prime mover or flywheel is in the engine house or not;
 - The headrace and tailrace of every water-wheel and water turbine;
 - Any part of a stock-bar which projects beyond the head stock of a lathe; and
 - Unless they are in such position or of such construction as to be safe to every person employed in the factory as they would be if they were securely fenced, the following, namely,
 - every part of an electric generator, a motor or rotary converter;
 - every part of transmission machinery, and
 - every dangerous part of any other machinery.

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CHAPTER IV SAFETY

Further precautions prescribed under sub-section (2) of section 21

* Further safety precautions :

- (1) Without prejudice to the provisions of sub-section (1) of section 21 in regard to the fencing of machines, the further precautions specified in the Schedules annexed hereto shall apply to the machines noted in each Schedule.
- (2) This Rule shall come into force in respect of any class or description of factories, where machines noted in the said Schedules are in use, on such date as the State Government may by notification in the Official Gazette, appoint in this behalf.

- SCHEDULE -I

(Textile Machinery except
Machinery & used in Jute Mills)

- SCHEDULE -II

(Cotton Ginning)

- SCHEDULE -III

(Wood Working Machinery)

- SCHEDULE -IV

(Rubber Mills)

- SCHEDULE -V

(Centrifugal Machines)

- SCHEDULE -VI

(Power Press)

- SCHEDULE -VII

(Shears, Slitters & Guillotine Machines)

* Register of specially trained adult worker :

Register of workers attending to machinery as provided in sub-section (1) of section 22 shall be in form no. 8

* Tight fitting clothing :

A worker required to wear tight fitting clothing under sub-section (1) of section 22 shall be provided by the occupier with such clothing which consist of at least a pair of closely fitting full pants and closely fitting half sleeves shirt or vest. Such clothing shall be returned to the occupier on termination of service or when new clothing is provided. Such workers shall be paid washing allowance which shall not be less than Rs. 10/- per month."

* **Rules prescribed under section 41 :**

Belt etc., to be regularly examined

All belts shall be regularly examined to ensure that the joints are safe and the belts are at proper tension.

Rules prescribed under sub-section (2) of section 23

Employment of young persons on dangerous machines :

The machines specified in section 28, 29, 30 and the following machines shall be deemed to be of such dangerous character that young persons shall not work on them unless the provisions of sub-section (1) of section 23 are complied with :

- (i) Power presses other than hydraulic presses.
- (ii) Milling machines used in the metal trades,
- (iii) Guillotine machines
- (iv) Circular saws
- (v) Platen printing machines

* **Rules prescribed under sub-section (1) of section 28**

Hoist examination - particulars of -

A register shall be maintained to record particulars of examination of hoists or lifts and shall give particulars as shown in Form No. 9

Exemption under sub-section (4) of section 28

Exemption of certain hoists and lifts

In pursuance of the provisions of sub-section (4) of section 28, in respect of any class or description of hoist or lift specified in the first column of the following schedule, the requirements of the section 28 specified in the second column of the said schedule and set opposite to the class of description of hoist or lift shall not apply.

I	II
Class or description of hoist or lift Hoists or lifts mainly used for raising Materials for charging blast furnaces or lime kilns. Hoists not connected with mechanical Power and which are not used for carrying Persons.	Requirements which shall not apply Sub-section (1) (b) in so far as it requires a gate at the bottom landing; sub-section (1) (d); sub-section (1) (e). Sub-section (1)(b) in so far as it requires the hoist way or enclosure to be so constructed to prevent any person or thing from being trapped between any part of the hoist or lift and any fixed structure; or moving part sub-section (1)(c).

* **Rules prescribed under sub-section (1) of section 29**

Lifting machines, chains, ropes and lifting tackles :

- (1) No lifting machine and no chain, rope or lifting tackle except a fiber rope or fiber rope sling shall be taken in use in any factory, for the first time therein unless it has been tested and all parts have been thoroughly examined by a competent person and a certificate of such test and examination specifying the safe working load or loads and signed by the person making the test and examination has been obtained and is kept available for inspection.
- (2) A register in Form 10 containing the particulars therein specified shall be kept for every examination made under sub-rule
 - (i) The register shall be readily available for inspection
- (3) (a) Every jib-crane so constructed that the safe working load may be varied by the raising or lowering of the jib, shall have attached thereto either an automatic indicator of safe working loads an automatic jib angle indicator and a table indicating the safe working loads at corresponding indication of the jib or corresponding radii of the load.
 - (b) A table showing the safe working load of every kind and size chain, rope or lifting tackle in use, and, in the case of a multiple sling, the safe working loads at different angles of the legs, shall be posted in the store room or place, where or in which the chains, ropes or lifting tackles are kept in prominent positions of the premises and no rope, chain or lifting tackle not shown in the table shall be used in a factory unless in the case of lifting tackle, the working load thereof, or in the case of a multiple sling, the safe working load at different angles of the legs, is plainly marked upon it.
- (4) All rails on which a traveling crane moves and every track on which the carriage of a transporter or runway moves, shall be of proper size and adequate strength and have an even running surface. Every such rail or track shall be properly laid and maintained and shall be adequately supported.
- (5) All chains and lifting tackle, except a rope sling, shall unless they have been subjected to such other heat treatment as may be approved by the State Government, be effectively annealed under the supervision of a competent person at the following intervals, namely
 - (i) All chains, slings, rings, hooks, shackles and swivels used in connection with molten metal or molten slag or when they are made of 12.7 milli meters bar or smaller, at least once in every six months.
 - (ii) All another chains, rings, hooks, shackles and swivels in general use at least once in every twelve months :

- Provided that chains and lifting tackle not frequent use shall subject to the approval necessary and particulars of such annealing shall be entered in a register in Form 10.
- (6) Nothing in sub-rule (5) shall apply to the following classes of chains and lifting tackle namely
- (i) Chains made of malleable cast iron.
 - (ii) Plate link chains.
 - (iii) Chains, rings, hooks, shackles and swivels made of steel or any non-ferrous metal.
 - (iv) Pitched chains, working on sprocket or pocketed wheels
 - (v) Rings, hooks, shackles and swivels permanently attached to pitched chains; pulleys blocks or weighing machines
 - (vi) Hooks and swivels having screw threaded parts or ball bearing or other case hardened parts.
 - (vii) Socket shackles secured to wire ropes by white metal capping.
 - (viii) Bordeaux connections.
 - (ix) Any chain of lifting tackle which has been subjected to the heat treatment known as "normalizing" instead of annealing. Such chains and lifting tackle shall be thoroughly examined by a competent person at least once in every twelve months, and particulars of such examination shall be entered in the register in Form 10.
- (7) All lifting machines, chains ropes and lifting tackle except a fiber rope or fiber sling, which have been lengthened, altered or repaired by welding or otherwise; shall, not be used again, unless it is adequately tested and examined by a competent person and certified in writing by him to be in order.
- (8) No person who has not completed eighteen years of age no adult who is not sufficiently trained in the working of lifting machines and acquainted with the hazards of the machines shall be employed as a driver of a lifting machine whether driver by mechanical power or otherwise, or to give signals to a driver.

*** Passage Ways For Cranes :**

- (a) To provide access to rail tracks of over head traveling cranes suitable passage ways of at least 50 cm width with toe boards and double hand rails 90 cm high shall be provided along side, and clear of the rail tracks of over head traveling cranes so that no moving part of the crane can strike persons on the way, and the passage way shall be at a lower level than the crane track itself, safe access ladders shall be provided at suitable intervals to afford access to these passage ways, and form passage ways to the rails tracks.
- (b) The state Government may in writing exempt any existing factory from the provisions of sub-rule (1) if the construction of the factory is such as to make it impossible to provide such a passage way.

* **Pressure Vessels or Plant :**

(1) **Interpretation :** In this rule

- (a) "Design pressure" means the maximum pressure that a pressure vessels or plant is designed to withstand safely when operating normally.
- (b) "Maximum permissible working pressure" means the maximum pressure at which a pressure vessel or plant is permitted to be operated or used under this rule and is determined by the technical requirement of the process.
- (c) "Plant" means a system of piping that is connected to a pressure vessel and so used to contain a gas vapour or liquid under pressure greater than the atmospheric pressure and includes the pressure vessels.
- (d) "Pressure vessel" means a vessel that may be used for containing, storing, and distributing, transferring, distilling, processing or otherwise handling any gas, vapour or liquid under pressure greater than the atmospheric pressure and includes any pipeline fitting or other equipment attached thereto or used in connection therewith.

(2) **Exception :** Nothing in this rule shall apply to

- (a) Vessels made of ferrous materials having an internal operating pressure not exceeding 1 Kilogram per square centimeter;
- (b) Steam boilers, steam and feed-pipes and their fitting coming under the purview of the Indian Boilers Act, 1923;
- (c) Metal bottles or cylinders used for storage or transport of compressed gases or liquefied or dissolved gases under pressure covered by the Gas Cylinder Rules, 1981 framed under the Indian Explosives Act, 1884
- (d) Vessels in which internal pressure is due solely to the static head of liquid
- (e) Vessels with a nominal water capacity not exceeding 500 liters connected in a system containing air that is compressing air that is compressed to serve as a cushion.
- (f) Vessels for unclear energy application;
- (g) Refrigeration that having a capacity of 3 tons or less of refrigeration in 24 hours; and
- (h) Working cylinders of steam engines or prime movers and steam traps; turbines; casings; compressor cylinders; steam separators or dryers; steam strainers; steam de-super-heaters; oil separators; air receivers for fire sprinkler installations; air receivers of monotypes machines provided maximum working pressure of the air receiver does not exceed 1.33 kilograms per square centimeter and the capacity of 15 liters; air receivers of electrical circuit breakers; air receivers of electrical relays; air vessels on pumps, pipe coils accessories of instruments and appliances such as cylinders and piston assemblies used for operating relays and interlocking type of

guards; vessels with liquids subjected to static head only; and hydraulically operating other than any cylinders communicating with an air loaded accumulator.

(3) Design and construction

Every pressure vessels or plant used in a factory -

- (a) Shall be properly designed on sound engineering practice;
- (b) Shall be of good construction, sound material, adequate strength and free from any patent defects; and;
- (c) Shall be properly maintained in a safe condition;

Provided that the pressure vessels or plant in respect of the design and construction of which there is in Indian standard or a standard of the country of manufacture or any other law or regulation in force, shall be designed and constructed in accordance with the said standard, law or regulation, as the case may be, and a certificate thereof shall be obtained from the manufacturer or from the competent person which shall be kept and produced on demand by an Inspector.

(4) Safety Devices

Every pressure vessel shall be fitted with-

- (a) A suitable safety valve or other effective pressure relieving device of adequate capacity to ensure that the maximum permissible working pressure of the pressure vessels shall not be exceeded. It shall be set to operate at a pressure not exceeding the maximum permissible working pressure and when more than one protective device is provided, only one of the devices need be set to operate at the maximum permissible working pressure and the additional device shall set to discharge at a pressure not more than 5 per cent in excess of the maximum permissible working pressure
- (b) A suitable pressure gauge with a dial range not less than 1.5 times the maximum permissible working pressure, easily visible and designed to show at all times the correct internal pressure and marked with a prominent red mark at the maximum permissible working pressure of the pressure vessel ;
- (c) A suitable nipple and end globe valve connected for the exclusive purpose of attaching a test pressure gauge for checking the accuracy of the pressure gauge referred to in clause (b) of this sub-rule;
- (d) A suitable stop or valves by which the pressure vessel may be isolated from other pressure vessel or plant or source of supply of pressure. Such a stop valve or valves shall be located as close to the pressure vessel as possible and shall be easily accessible ;
- (e) A suitable drain cock or valve at the lowest part of the pressure vessel for the discharge of the liquid or other substances that may collect in the pressure vessel :
Provided that it shall be sufficient for the purpose of this sub-rule if the safety valve

or pressure relieving device, the pressure gauge and the stop valve are mounted on a pipe line immediately adjacent to the pressure vessel and where there is a range of two or more similar pressure vessel served by the same pressure lead, only set of such mounting need be fitted on the pressure lead immediately adjacent to the range of pressure vessels provided they can not be isolated.

(5) Pressure reducing devices

- (a) Every pressure vessel which is designed for a working pressure less than the pressure at the source of supply; or less than the pressure which can be obtained in the pipe connecting the pressure vessel with any other source of supply, shall be fitted with a suitable pressure reducing valve or other suitable automatic device to prevent the maximum permissible working pressure of the pressure vessel exceeded.
- (b) To further protect the pressure vessel in the event of failure of the reducing valve or device, at least one safety valve having a capacity sufficient to release all the steam, vapour or gas without undue pressure rise as determined by the pressure at the source of supply and the size of the pipe connecting the source of supply shall be fitted on the low pressure side of the reducing valve.

(6) Pressure vessel or plant being taken into use

- (a) No new pressure vessel or plant shall be taken into use in a factory after coming into force of this rule unless it has been hydrostatically tested by a competent person at a pressure at least 1.3 times the designed pressure, and no pressure vessel or plant which has been previously used or has remained isolated or idle for a period exceeding 2 months or which has undergone alterations or repairs shall be taken into use in a factory unless it has been thoroughly examined by a competent person externally and internally if practicable and has been hydrostatically tested by the competent person at a pressure which shall be 1.5 times the maximum permissible working pressure;

Provided however, that the pressure vessel or plant which is so designed and constructed that it can not be safely filled with water or liquid or is used in service when ever some traces of water can not be tolerated, shall be pneumatically tested at a pressure not less than the design pressure or the maximum permissible working pressure as the case may be :

Provided further that the pressure vessel or plant is lined with glass shall be tested hydrostatically or pneumatically as required at a pressure not less than the design pressure or maximum permissible working pressure as the case may be.

- (b) No pressure vessel or plant shall be used in a factory unless there has been obtained from the make of the pressure vessel or plant or from the competent person a certificate specifying the design pressure or maximum permissible working pressure thereof and stating the nature of tests to which the pressure vessel or plant and its fittings (if any) have been subjected, and every pressure vessel or plant so used in

a factory shall be marked so as to enable it to be identified as to be the pressure vessel or plant to which the certificate relates and the certificate shall be kept available for perusal by the Inspector.

- (c) No pressure vessel or plant shall be permitted to be operated or used at a pressure higher than its design pressure, or the maximum permissible working pressure as shown in the certificates.

(7) In-service test and examinations

- (a) Every pressure vessel or in service shall be thoroughly examined by a competent person

- (i) Externally, once in every period of six months;

- (ii) Internally, once in every period of twelve months;

Provided that if by reason of the construction of a pressure vessel or plant, a through internal examinations not possible, this examination may be replaced by a hydrostatic test which shall be carried out once in every period of two years :

Provided further that for a pressure vessel or plant in continuous process which cannot be frequently opened, the period of internal examination may be extended to four years and

- (iii) Hydrostatically tested once in every period of four years :

Provided that in respect of pressure vessel or plant with thin walls such as sizing cylinder made of copper or any other non-ferrous metal, periodic test may be dispensed with subject to the condition that the requirements laid down in sub-rule (8) are fulfilled :

Provided further that when it is impracticable to carry out through external examination of any pressure vessel or plant every six months as required in sub-clause (i) of this clause or if owing to its construction and use a pressure vessel or plant cannot be hydrostatically tested as required in sub-clause (ii) and (iii) of this clause, through external examination of the pressure vessel or plant shall be carried out at least once in every period of two years and at least once in every period of four years a through systematic nondestructive test like ultrasonic test for metal thickness or other defects of all parts of the failure of which might lead to eventual rupture of the pressure vessel or plant shall be carried out.

- (b) The pressure for the hydrostatic test to be carried out for the purpose of this sub-rule shall be 1-24 times the design pressure or 5 times the maximum permissible working pressure whichever is less.

(8) Thin walled pressure vessel or plant

- (a) In respect of any pressure vessel or plant of thin walls such as sizing cylinder made of copper or any other non-ferrous metal, the maximum permissible working pressure for every year of use after the first five years and no such cylinder shall be

allowed to continue to be used for more than twenty years after it was first taken into use.

(b) If any information as to the date of construction, thickness of walls or maximum permissible working pressure is not available the age of such pressure vessel or plant shall be determined by the competent person in consultation with the Chief Inspector from the other particulars available with the manager.

(c) Every new and second hand pressure vessel or plant of walls to which repair is likely to effect its strength or safety have been carried out, shall be tested before used to at least 1.5 times its maximum permissible working pressure.

(9) Report by competent person

(a) If during any examination any doubt arises as to the ability to the pressure vessel or plant to work safely until the next prescribed examination, the competent person shall enter in the prescribed register his observations and conclusions with other relevant remarks with reasons and may authorize the pressure vessel or plant to be used and kept in operation subject to a lowering of maximum permissible working pressure, or to more frequent or special examination or test or subject to both of these conditions.

(b) A report of every examination or test carried out shall be completed in Form No. 11 and shall be signed by the person making the examination or test and shall be kept available for perusal by the Inspector at all hours when the factory or any part thereof is working.

(c) Where the report of any examination under this rule specifies any conditions for securing the safe working of any pressure vessel or plant the pressure vessel or plant or any part thereof continue to be used with safety unless certain repairs are carried out or unless any other safety measure is taken.

(d) The competent person making report of any examination under this rule shall within seven days of the completion of the examination send to the inspector a copy of the report in every case where the maximum permissible working pressure is reduced or the examination shows that the pressure vessel or plant or any part thereof can not continue to be used with safety unless certain repairs are carried out or unless any other safety measures is taken.

(10) Application of other laws

(a) The requirements of this rule shall be in addition to and without any prejudice to and not in derogation of the requirements of any other law in force.

(b) Certificate or reports of any examination or test of any pressure vessel or plant to which sub-rule (7) to (9) do not apply, conducted or required to be conducted under any other law in force and other relevant record relating to such pressure vessel or plant, shall be properly maintained as required under the said law and shall be produced on demand by the Inspector.

*** Safety of Water Sealed Gas-holder :**

- (1) The expression "gas-holder" in this rule means a water-sealed gasholder which has a storage capacity of not less than 141.5 cubic meters (5,000 cu. ft.).
- (2) Every-gas-holder shall be of good construction, sound material and adequate strength and shall be properly maintained.
- (3) Where there is more than one gasholder in a factory, every gasholder shall be marked in a conspicuous position with a distinguishing number of letters.
- (4) Every gasholder shall be thoroughly examined externally by a competent person at least once in a period of 12 months.
- (5) In the case of a gas-holder of which any lift has been in use for more than 10 years, the internal state of sheeting shall, within one year of the coming into operation of this rule and thereafter at least every period of four years, be examined by a competent person by means of electronic or other accurate devices :
Provided, that if the Chief Inspector is satisfied that such electronic or other accurate devices are not available, he may permit the cutting of samples from the crown and the sides of the holder;

Provided further that if such examination raises a doubt as to the soundness of the internal state of the sheeting an internal visual examination shall be made.

- (6) All possible steps shall be taken to prevent or minimize ingress of impurities in the gasholders.
- (7) No gasholder shall be repaired or demolished except under the direct supervision of a person who by his training and experience and his knowledge of the necessary precautions against risks of explosion and or persons being overcome by gas, is competent to supervise such work.
- (8) (i) All Sample discs cut under sub-rule (5) shall be kept readily available for inspection.
(ii) A permanent register duly signed by the occupier or manager shall be maintained giving the following particulars :-
 - (a) The distinguished number of letter of gas-holder marked thereon under sub-rule (3) and the particulars or manufacturer i.e., maker's name, date of manufacture, capacity, number of lifts, and pressure thrown by holder when full of gas;
 - (b) The dates of examination carried out as required under sub-rules (4) and (5) above and by whom carried out;
 - (c) The methods of examination used;
 - (d) Date of painting;
 - (e) Nature of repairs and names of persons carrying out repairs; and
 - (f) Remarks

- (i) The results of examinations by a competent person under sub-rules (4) and (5) shall be in Form No. 11-A.
- (ii) A copy of the report in Form No. 11-A shall be kept in the register and both register and the report shall be readily available for inspection.

* **Reaction Vessels and Kettles :**

- (1) This rule applies to reaction vessels and kettles (herein after referred to as "reaction vessels" which normally work at the pressure not above the atmosphere pressure but in which there is likelihood of pressure being created above the atmosphere pressure due to reaction getting out of control or any other circumstances.
- (2) In the event of the vessel being heated by electrical means, a suitable control device shall be provided to prevent the temperature exceeding the safe limit.
- (3) Where steam is used for heating purposes in reaction vessel, it shall be supplied though a suitable automatic device to prevent the maximum permissible steam pressure being exceeded unless the pressure of the steam in the supply line itself can not exceed the said maximum permissible pressure.
- (4) A suitable safety valve require disc, of adequate size and capacity shall be provided to effectively prevent the pressure being built up in the reaction vessel beyond the safe limit. Effective arrangement shall be made to ensure that the released gases, fumes, vapors, liquids or dusts, as the case may be, are lead away and disposed or through suitable pipes without causing any hazard flammable gases or vapour are like to be vented out from the vessel the discharge end shall be provided with a flame arrestor.
- (5) Every reaction vessel shall be provided with a pressure gauge having an appropriate range.
- (6) In addition to the device as mentioned in the above clauses, means shall be provided for automatically stopping the feed into the vessel as soon as process condition from the normal limit to an extent which can be considered as dangerous.
- (7) Where necessary, an effective system for cooling, flooring or blanketing shall be provided for the purpose of controlling the reaction and process conditions within the safe limits of temperature and pressures.
- (8) Automatic auditory and visual warning devices shall be provided for clear warning whenever process conditions exceed the present limits. The device wherever possible shall be integrated automatic process correction system.
- (9) A notice pointing out the possible circumstances in which pressure above atmospheric pressure may be built up in the reaction vessel, the dangers, involved and the precautions to be taken by the operators shall be displayed at the conspicuous place near the vessel.

Rules prescribed under sub-section (2) of section 34 :

* Excessive Weights :

- (1) No woman or young person shall unaided by another person, lift, carry or move by hand on head, any material, article, tool or appliance exceeding the maximum limit in weight set out in the following schedule :

SCHEDULE I	
Persons	Maximum weight of Material article total by Appliance
(a) Adult male	55.0 Kilograms
(b) Adult female	29.5 Kilograms
(c) Adolescent male	29.5 Kilograms
(d) Adolescent female	20.4 Kilograms
(e) Male child	15.9 Kilograms
(f) Female child	13.6 Kilograms

- (2) No woman or young person shall engage, in conjunction with others in lifting, carrying or moving by hand or on head, any material article, tool or appliance if the weight thereof exceed the lowest weight fixed by the schedule to sub-rule (1) for any of the persons engaged, multiplied by the number of the persons engaged.

5.4 Safety Colour Code :

The safety colour code is used to prepare notices, signs and labels. The attention can easily be diverted towards hazards easily hence the accidents are reduces. The safety colour codes are kept and used as follows :

- (1) General colour code is used to indicate hazardous places, fire fighting equipments, doors and traffic lines.
- (2) Special colour code is used for gas cylinder & gas pipe line
- (3) Separate colour code is for gang ways in a workshop for trace passing.
- (4) Colour code giving good mental effect is used for walls and ceiling.

By indentifying the locations using proper colour code, the informations of hazards associated with them is provided. Only by visualization of colour code used the idea towards the hazard is occurring in the mind. If the colour code is used to develop proper thinking then by avoiding the improper methods in work, the safety can be brought by preventing the dangerous conditions. When safety is gained, the accidents are automatically reduces can be said as contribution of colour code. Proper colours are useful to make the atmosphere of the workplace in the industry of a type which likes the workers, therefore, they feel ease and feel at home during their working. Due that the spoilage of mental condition stops and accidents can be prevented.

5.4.1 Yellow colour :

The orange-yellow colour is used to indicate hazards like to show the places to be covered by the guards, therefore, if there is no guard at that place can be immediately realised & known. This colour is also used to indicate the part or portion where the electric current or temperature have their maximum values. Signs for caution and warning signs and columns are shown by using the yellow and black colours alternatively in the forms of strips of sufficient width. The colour for surrounding to these strips is used to get desired contrast.

5.4.2 Red colour :

Stop symbols, emergency stop devices, fire fighting equipments, operating controls, tanks and cylinders containing harmful materials, inner sides of machine guards and internal dangerous portions are painted by red-colour. Hence it can be said that the signs & signals indicating danger are to be painted using red colour code for them.

5.4.3 Green colour :

The green colour code is used for the ways & passage of escape. First aid station, Go traffic signal and safety installations. The green colour is therefore used to show the normal and safe conditions of mechanisms, circuits and parameters like temperature & pressure.

5.4.4 Blue colour :

When it is not possible to use yellow, red and green colour at a particular location, then colour code is used at that places for signs, signals notices giving informations.

5.4.5 White colour :

The white achromatic colour is used to indicate or check the voltage, operating speed and other conditions which cannot be automatically controlled. Way for trace-passing in traffic and scrap disposal are shown by white lines and strips.

5.5 Fire prevention and fire fighting methods :

5.5.1 Causes of fire :

1. Handle the materials which can catch fire with or without care.
2. Loose connections of cables and wires.
3. Habit of smoking and cigarettes.
4. Insufficient lubrication of fast moving machine parts.
5. Short circuiting of electrical wirings.
6. Using the conductors beyond their current capacity.
7. Sparking of electric switch & starters.
8. Materials which can burn like paper, card-board, pieces of wood & oily clothes & cotton waste are kept anywhere on the floor.
9. Electrical equipments are not maintained as per rules.

10. Sparks generated in electric arc welding.
11. Due to atmospheric discharge.
12. Insufficient construction of temporary heating.

5.5.2 Classification of materials for fire :

1. Class-A fire → Wood, paper etc. of which the fire extinguished by water.
2. Class-B fire → Blanketing is done to extinguish the fire extinguishing the fire of oil, solvent and petroleum products.
3. Class-C fire → Pressurised gases of which the fire is extinguished using inert gas or powder by decreasing the pressure of burnt gases.
4. Class-D fire → Special mediums & technics are used to extinguish the fire of aluminium, magnesium, zinc and potassium.
5. Class-E fire → Electrical non-conductive materials are selected for fire fighting of the transformers, switch board & motors.

5.5.3 Fire fighting methods :

1. Class-A fire → For fire fighting in this case portable type foam & sodaacid equipments, water pump and gas cartridge are utilised.
 2. Class-B fire → For fire fighting vapourised liquid foam, carbon dioxide and portable equipments of chemical spray are used.
 3. Class-C fire → Portable equipments for dry chemical powder & carbon dioxide are sprayed as fire fighting material to extinguish the fire.
 4. Class-D fire → The dry materials are sprayed with the help of fire fighting equipments to extinguish the fire.
 5. Class-E fire → Fire fighting materials such as carbon dioxide, dry chemicals, and vaporising liquid is sprayed using the portable equipments for extinguishing this type of class-E fire.
- Class-A fire can be extinguished by water and solution containing more water.
- For extinguishing the class-B fire it is necessary to burning materials & hence water is not used in it because the fire spreads due to the use of water.
- The class-C fire of burning gases can be extinguished using inert gases. To slow down the fire dry powder can also be used for such materials.
- The class-D fire of burning metals can be extinguished using special mediums and techniques. As metal reacts with the water, it is not suitable for this type of fire.
- The class-E of the materials can be extinguished using electrical insulating materials to avoid the shock to the persons working for fire fighting.